

Enterprise Architecture Standards — Knowledge Base

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This KB defines the enterprise-level architecture standards for NTUC First Campus (NFC). All design, development, and support agents **MUST** ground their decisions in these standards.

EA1: Organisation Overview

Attribute	Details
Organisation	NTUC First Campus Limited (NFC)
Established	1977
Domain	Early childhood education and services
Headquarters	16 Raffles Quay, #23-01, Hong Leong Building, Singapore 048581
Staff Strength	7,000+
Centres	180+ childcare centres across Singapore
UEN	202308481C

Business Units

Unit	Type	Centres	Focus
MyFirstSkool (MFS)	Anchor Operator	160+	Childcare services
Little Skool House (LSH)	Partner Operator	14	Childcare services
ChangeMakers Explorer	Premium Operator	6	Childcare services
SEED Institute	Enrichment arm	—	Afterschool student care, enrichment programmes, outdoor adventure camps

EA2: Technology Stack

Layer	Technology	Purpose	Status
Backend (SN2)	Go (Golang)	Core backend language	Mandatory
Frontend (SN2)	React	Web application	Mandatory

Layer	Technology	Purpose	Status
Mobile (SN2)	React Native	Cross-platform mobile app	Mandatory
Unit Testing	Jest	JavaScript/React testing	Mandatory
Code Quality	SonarCloud	Static analysis	Mandatory
Database (Relational)	MySQL	Primary relational store	Mandatory
Database (NoSQL)	MongoDB	Document store	Mandatory
Data Analytics	BigQuery	Analytics and reporting	Mandatory
CI/CD	Bitbucket Pipelines	Backend CI/CD	Mandatory
CI/CD (Mobile)	Bitrise	Mobile app CI/CD	Mandatory
Source Control	Bitbucket	Git repository hosting	Mandatory
Observability	OpenTelemetry	Distributed tracing	Mandatory
Monitoring	Datadog	Infrastructure & app monitoring	Mandatory
Monitoring	Cron Jobs	Scheduled job monitoring (SN2)	Mandatory
Monitoring	GCP Cloud Native Logging	Cloud logging	Mandatory
Feature Flags	Split.io	Feature flag management	Mandatory
Communications	Twilio	SMS/Voice	Mandatory
Email	SendGrid	Transactional email	Mandatory
Push Notifications	Firebase Cloud Messaging	Mobile push	Mandatory
Chat	Sendbird	In-app messaging	Mandatory
Live Chat (Support)	Freshchat	Customer support chat	Mandatory
Ticketing	ServiceNow	ITSM / incident management	Mandatory
Ticketing (Dev)	JIRA	Development tracking	Mandatory
ITSM	ServiceNow	Primary IT service management	Mandatory
RPA	UiPath	Robotic process automation (3 attended + 1 dev license)	Mandatory
HR Management	SuccessFactors	Employee lifecycle management	Mandatory
CRM / Marketing	HubSpot	Marketing & Sales Hub	Mandatory
CRM Middleware	Middle Stage	Custom integration middleware	Mandatory

Layer	Technology	Purpose	Status
Google Workspace	Google Admin Console	Enterprise collaboration	Mandatory
Google Automation	GAM API	Advanced Google administration	Mandatory
Test Automation (UI)	TestSigma	UI test automation	Mandatory
Performance Testing	JMeter	Load and performance testing	Mandatory

Infrastructure Stack (Multi-Cloud)

Layer	GCP	AWS	Purpose
Compute	GCE, GKE	EC2, EKS	Virtual machines, Kubernetes
Storage	Cloud Storage (GCS)	S3, Glacier	Object storage, archival
Block Storage	Persistent Disk	EBS	VM-attached storage
File Storage	Filestore	EFS	Shared file systems
Database (Relational)	Cloud SQL	RDS (PostgreSQL/MySQL)	Managed databases
Database (NoSQL)	Firestore	DynamoDB	Document databases
Database (Wide-column)	Bigtable	—	Time-series, analytics
Cache	Memorystore (Redis)	ElastiCache (Redis)	In-memory caching
Networking	Shared VPC, Cloud Load Balancing	ELB (ALB/NLB)	VPC, load balancing
DNS	Cloud DNS	Route53	Domain name resolution
CDN	Cloud CDN	CloudFront	Content delivery
Connectivity	Cloud Interconnect	Direct Connect	Dedicated connectivity
IaC	Terraform (state in GCS)	Terraform (state in S3)	Infrastructure as Code
Image Management	Packer	Packer	Golden images
Config Management	Ansible	Ansible	Server configuration
CI/CD	Jenkins, GitHub Actions	—	Build and deploy pipelines
GitOps	ArgoCD / Flux	—	Kubernetes deployments
Container Registry	Artifact Registry	ECR	Docker image storage
Observability	Google Cloud Operations (Stackdriver)	—	Cloud monitoring

Layer	GCP	AWS	Purpose
Metrics	Prometheus, Grafana	—	Metrics and dashboards
Logging	ELK Stack / Fluentd	—	Centralised logging
Alerting	PagerDuty / OpsGenie	—	Incident alerting
Security (WAF)	Cloud Armor	AWS WAF	Web application firewall
Security (IAM)	Cloud IAM	IAM	Identity & access
Security (Keys)	KMS	KMS	Key management
Security (Secrets)	Secret Manager	—	Secrets storage
Security (Posture)	Security Command Center	Security Hub	Security monitoring
FinOps	GCP Billing & Cost Management	AWS Cost Explorer, Trusted Advisor	Cost optimisation
Identity	On-premises Active Directory (GCP-hosted), Microsoft Entra ID	—	Identity management
PKI	Certificate Authority (AD CS)	—	Certificate management

MANDATORY RULE: Core enterprise applications (including SN2) are hosted on GCP. AWS is used primarily for storage services and DNS. Any architectural change must maintain this separation.

EA3: Core Application — SkoolNet 2 (SN2)

Purpose

End-to-end preschool management system — "single source of truth" for preschool data.

Capabilities

- Automates enrolment planning, attendance, and vacancy tracking
- Online parent registration with integrated payments (PayNow)
- Child safety via digital check-in/out
- Direct integration with ECDA (regulatory body) for subsidies and reporting
- Data accuracy and consistency across all business units

Users

User Type	Count	Interface	Key Actions
Parents	~44,000+	Parent App/Portal	Registration, fee payments, child portfolios
Children (served)	~28,000	—	—
Guardians	~29,000+	Guardian App	Child attendance check-in/out
Teachers	5,000+	Staff App	Classroom management, attendance, learning observations
Principals	180+	Staff Portal	Centre administration, enrolment management, event planning
Head Office	—	Staff Portal	Strategic planning, financial auditing, data analytics

Development Statistics (2025)

Metric	Value
Release Cycle	Monthly
Sprint Duration	2 weeks
Number of Squads	3
FTE Capacity/Month	180 man-days (9 headcounts × 20 days)
Contractor Capacity/Month	400 man-days (20 headcounts × 20 days)
Feature Releases/Month	13
Bug Fixes/Month	8

L2 Support Statistics

Metric	Value
Ticket Volume	260–300 tickets/month (baseline)
Support Hours	Mon-Fri, 08:30–18:00 SGT (excl. public holidays)
Peak Periods	Weekdays 7–9 AM & 5–7 PM; Oct-Dec (new enrolments); Monday after releases
Team Size	8 resources (2 Team Leads, 6 Support Engineers)

SLA (L2 Support)

Severity	Response	Resolution	Definition
Blocker	30 min	4 hours	System blocker, no workaround
High	30 min	8 hours	System blocker with workaround
Medium	1 hour	3 working days	Intermittent / non-critical
Low	2 hours	7 working days	Cosmetic / minor bugs

EA4: Architecture Patterns

Backend: Go (Golang) Microservices

- Core backend built in Go
- REST APIs for client-server communication
- MySQL for relational data, MongoDB for document storage
- BigQuery for analytics workloads
- OpenTelemetry for distributed tracing

Frontend: React Web + React Native Mobile

- React for web portal (Staff Portal, Parent Portal)
- React Native for cross-platform mobile (Parent App, Guardian App, Staff App)
- Jest for unit testing, SonarCloud for quality gates

Infrastructure: Multi-Cloud (GCP Primary, AWS Secondary)

- GCP hosts core applications, data & analytics, Active Directory
- AWS for storage (S3), DNS (Route53), archival (Glacier)
- Kubernetes (GKE/EKS) for container orchestration
- Terraform (IaC) for all infrastructure provisioning
- GitOps (ArgoCD/Flux) for Kubernetes deployments

Integration Patterns

- Twilio / SendGrid for communications
- Firebase Cloud Messaging for mobile push

- Sendbird for in-app messaging
- Freshchat for live customer support
- PayNow integration for payment processing
- ECDA integration for regulatory compliance

EA5: Service Domains

Area	Domain	Description
A	SN2 Application Development & Support	Feature development, bug fixes, L2 technical support
B	Quality Assurance	Manual testing, automation (TestSigma), API testing, performance testing (JMeter)
C	Level 1 Help Desk	First-contact triage for internal staff and external parents
D	Google Workspace Support	Administration, security, governance of Google Workspace
E	Infrastructure Operations	CloudOps, FinOps, SecOps across GCP/AWS multi-cloud
F	Robotic Process Automation	UiPath-based automation (25 existing use cases)
G	Professional Services for SaaS	SuccessFactors (HR) and HubSpot (CRM/Marketing) enhancements

EA6: Support Model

L1 Help Desk

Metric	Value
Internal Users	7,000+ staff (HQ + centres); ~40% Chinese-speaking educators
External Users	~44,000+ parents
Ticket Volume	~1,100 tickets/month
Support Hours	Mon-Fri, 07:00–19:00 SGT
Channels	ServiceNow (ITSM), Email (auto-ticket), Phone, Freshchat (live chat)
Languages	Primary: English; Secondary: Chinese

Metric	Value
Peak Periods	Weekdays 7–9 AM & 5–7 PM; 1st of month; Monday after release

L1 SLA

Priority	Response	Escalation to L2/L3	Resolution	Adherence
S1	20 min	60 min	4 hours	90%
S2	30 min	90 min	8 hours	90%
S3	1 hour	1 working day	3 working days	90%
S4	2 hours	2 working days	7 working days	90%

Infrastructure SLA

Severity	Response	Resolution	Adherence
S1 (Critical)	15 min	4 hours	99%
S2 (High)	30 min	48 hours	95%
S3 (Normal)	1 hour	72 hours	95%
S4 (Low)	2 hours	120 hours	95%

EA7: Quality Assurance Standards

Metric	Value
UI Automation Test Cases (SN2)	2,200
API Tests	1,500
Regression Scripts	1,600
Bug Fixes/Month	8
Test Automation Tool	TestSigma (UI)
Performance Testing Tool	JMeter
Strategy Direction	"Automation First"

Quality Gates

- Feature testing, regression testing, performance testing, sanity testing at every release
- Automated tests integrated into CI/CD pipelines
- Defect warranty: vendor bears cost of Severity 1 & 2 defects found in UAT or within 30-day

EA8: Security & Compliance

Identity & Access Management

- On-premises Active Directory (hosted on GCP)
- Microsoft Entra ID (cloud identity)
- AD Connect for synchronisation
- Single Sign-On (SSO) for enterprise and third-party apps
- Conditional Access with MFA
- Least privilege enforcement

Cloud Security (SecOps)

- Cloud Armor (GCP) / AWS WAF for web application protection
- GCP Security Command Center / AWS Security Hub for posture management
- KMS for encryption at rest
- TLS for encryption in transit
- Secret Manager for secrets storage
- IAM Role/Policy enforcement, Service Account management
- Network security: Security Groups, Firewall Rules, VPN tunnels

Certificate Management (PKI)

- Internal Certificate Authority (AD CS)
- Certificate issuance, renewal, and revocation
- Validity monitoring and expiry alerting
- Template and trust chain management

Compliance

- Integration with ECDA for regulatory reporting (early childhood)
 - Audit support for infrastructure and identity management
 - SLA adherence tracking and reporting
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EA9: Infrastructure Scale

Metric	Value
Total Cloud Resources	~830+
Primary Cloud	Google Cloud Platform (GCP)
Secondary Cloud	Amazon Web Services (AWS)
Migration Status	Full migration from on-premise completed 2024
Uptime Target	99.9%
IaC Coverage	Terraform
Container Orchestration	Kubernetes (GKE on GCP, EKS on AWS)

FinOps

- Project-level budget alerts to prevent bill shock
 - Billing anomaly monitoring for spend spikes
 - Committed Use Discounts (CUDs) management based on historical spend
 - AWS Cost Explorer and Trusted Advisor for AWS spend
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EA10: Challenges & Technical Debt

Area	Challenges
SN2 Development	Outdated backend/frontend libraries; unpatched vulnerabilities; insufficient test automation; legacy complex codebase; lack of code quality ownership; fragmented end-to-end design knowledge
SN2 L2 Support	Burst volumes above baseline; recurring issues needing permanent fixes; knowledge retention; ticket bouncing between L1/L2/L3; L2 lacks code-level RCA skills
QA	Reliance on manual testing slows releases; inconsistent test data; slow automation-first transition; automated tests not integrated into CI/CD
Help Desk	Language coverage (40% Chinese educators); channel fragmentation; ServiceNow maintenance
Google Workspace	Reactive troubleshooting; delays in complex/high-risk resolution; increasing security/compliance demands
Infrastructure	Fragmented multi-cloud knowledge; manual processes; scaling team up/down

EA11: Future Roadmap

Technology Refresh (Planned Replacements)

Current System	Purpose	Status
Sage 300/Accpac	Financial management	To be replaced
Sage eProcurement	Procurement	To be replaced
Anaplan	Planning & budgeting	To be replaced
PageUp	Application Tracking System	To be replaced
SuccessFactors	HR (Succession, Performance, Learning, Employee Central, Payroll)	To be replaced

Strategic Direction

- AI/ML POCs and MVPs leveraging new AI capabilities
- SN2 roadmap: feature enhancements + regulatory compliance
- Transition from manual to "Automation First" in QA
- Self-service features to reduce L2 support reliance
- RPA expansion beyond 25 current use cases
- Real-time alerting for Google Workspace admin actions
- Administrative role reassessment and PAM controls

EA12: Operating Model Expectations

Strategic Partnership Model

NFC seeks a single strategic technology partner operating as an extension of the internal technology team:

Principle	Description
Outcomes-driven	Focused on service reliability, platform modernisation, delivery acceleration
Single partner	Coordinated delivery across all 7 service domains
Scalable	Scale team up/down based on business demands
Governed	Strong governance, security, and SLA adherence

Principle	Description
Delivery Manager	Partner provides single point of contact removing management burden
Knowledge continuity	Reduce dependency on individual resources
Modern tooling	Proactively identify efficiency gains through modern tools
Talent pool	Support new technologies: AI/ML, Low Code Platforms

Delivery Structure

Category	Services
Day 1 Services	SN2 Engineering (A), QA (B), RPA (F), Professional Services for SaaS (G)
Day 2 Services	SN2 L2 Support (A), L1 Helpdesk (C), Google Workspace (D), Infrastructure (E)

Current Operating Models

Area	Model	Composition
SN2 Development	Staff augmentation (near-term), Managed service (long-term)	Mix of FTEs (9) and contractors (20); internal Product Owner retained
SN2 L2 Support	Managed service	Incumbent vendor, 8 resources (2 TLs + 6 engineers), SLA-driven
L1 Help Desk	Outsourced managed service	Vendor-operated from 2 offshore locations, SLA-driven on volumetrics
Google Workspace	Internal SME (to be transitioned)	Currently small internal team; transitioning to strategic partner
Infrastructure	Managed service	Dedicated partner team for monitoring, security hardening, SLA adherence
RPA	Single contract resource	1 resource supporting 25 use cases + new implementations
QA	Hybrid workforce	Mix of manual testers + automation engineers embedded in squads

Delivery Composition Expectations

- Partner proposes optimal onsite-offshore mix per service area

- Shared vs dedicated resources based on domain complexity
- Scale up/down flexibility based on business demand
- Delivery Manager as single point of contact (partner-provided)
- Periodic performance reviews with replacement capability